

Model question for unit test

1. Explain the difference between Object-Oriented Analysis (OOA) and Object-Oriented Design (OOD) with suitable examples.
2. Why is use case description important? Discuss the important sections of a use case description with an example.
3. Discuss the different phases of Unified Process / Rational Unified Process (RUP) and their role in software development.
4. What are use cases and how can they be used for identifying and recording system requirements? Differentiate between include and extend relationships between use cases.
5. What are the various formats used to represent a use case? Explain the Fully-Dressed format with an example.
6. Explain the importance of the object oriented analysis in identifying conceptual classes and other elements and to construct domain model with scenario.
7. Explain data abstraction and behavioral abstraction with suitable examples. How are object responsibilities different from object behavior?
8. Create a use case diagram and activity diagram for an ATM (Automated Teller Machine) system. Consider both Bank Customers and Bank Staff as actors. Include all essential banking operations such as withdrawals, deposits, balance inquiries, and PIN changes. Also include maintenance operations performed by bank staff. Show appropriate include and extend relationships, and clearly indicate which use cases require authentication.
9. Create a use case diagram and activity diagram for a University Library Management System. The system should cater to Students, Faculty, Librarians, and System Administrators. Include functionality for book borrowing, returns, reservations, fine calculations, and catalog management. Clearly show relationships between different use cases and identify where authentication is required.
10. Create a use case diagram and activity diagram for an Online Food Delivery System. The system should handle customer orders, restaurant management, and delivery personnel. Consider the following actors: Customers, Restaurant Owners, Delivery Partners, and System Admin. Include essential use cases for order placement, payment processing, delivery tracking, and restaurant menu management. Include appropriate relationships (include, extend) where necessary.
11. Write short note on :
 - a) Functional requirements vs non-functional requirements
 - b) Three ways to applying UML
 - c) Reverse engineering
 - d) UML multiple perspectives
12. What is the purpose of developing conceptual model? How a domain model is is developed? Explain its component.